

Inside the Black Box. Simondon's Politics of Technology

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In 1923, Paul Valéry created an artificial world of antiquity. In it the sea could wash up things that, because of their brilliance, their hardness, and their unfamiliar form, interrupted and irritated well-established habits of thought. Nature or art? Given or created? Earthly or heavenly? *Eupalinos*, the architect, does not find himself in the position to decide. He throws back into the sea the shiny, ball-like thing that he had picked up from the shore only seconds before.¹

In the 1950s, the situation has changed remarkably. The Parisian consumer society uses the polished floors of exhibition halls and sales facilities in order to create encounters with similarly enigmatic and wonderful objects. However, one can no longer take these objects into one's hand nor throw them away. Things insist, as do the questions, for both department stores and commercial fairs visitors as well as for the audience attracted by the machine show that accompanied the Paris Cybernetics congress in 1951.²

Roland Barthes has brilliantly characterized the features of this thing-experience. In October 1955, the Citroën *D.S. 19* was presented to a large public at the occasion of the

¹ Paul Valéry, *Eupalinos. Or, the Architect*, transl. and with a preface by William McCausland, Stewart, London: Oxford University Press, 1932.

² The context for this congress was provided by Nobert Wiener's lectureship at the Collège de France. See Pierre de Latil, *Thinking by Machine. A Study of Cybernetics*, transl. by Y. M. Golla, London: Sidgwick and Jackson, 1956, pp. 3-24.

Parisian car show “Salon de l’Automobil.” During the evening of the first day of the show, Citroën’s sales managers counted a sensational 12,000 orders for the product. Two years later, Barthes published in *Mythologies* his compelling analysis of “The New Citroën.” It reads like an early summary of the process leading from the initial presentation of the futuristic vehicle to its massive distribution. Boiled down to a formula, Barthes’s conclusion is the following: first a gothic cathedral, then a kitchen.³

According to Barthes, the cultural appropriation of the *D.S.* (a pun on “*déesse*,” i.e. “Goddess”) corresponds to a transition from admiring a magical thing to using a mere apparatus, from seeing something to touching it, and from the outer skin to the interior of the technical object. As a result, a somewhat uncanny “alchemy of speed” is brought down to the familiar principle of *gourmandise*, “a relish in driving.” What, at first sight, seemed as a quasi-sacred object – a thing “descended from another universe” – transforms itself in the very moment one eventually sits behind the steering wheel into a cozy object of daily use: “The dashboard looks more like the working surface of a modern kitchen than the control-room of a factory.”⁴

In Barthes eyes, this is not an arbitrary move. It might only take a quarter of an hour, but the work of mediation, of “exorcism” – with Agamben, we might even say “profanation”⁵ – that here takes place with respect to a car does aim at a long-term goal. With telling ambivalence, Barthes describes this goal as “petit-bourgeois advancement,” *la promotion petite-bourgeoise*.⁶

However, it only seems as if the appropriating move from outside to interior implied a process of getting closer to the technical object. In fact, its inner side – the seats, the dash

³ Roland Barthes, “The New Citroën,” in: id., *Mythologies*, transl. by Annette Lavers, London: Cape, 1972, pp.88-90.

⁴ Ibid., p.89.

⁵ Giorgio Agamben, *Profanations*, transl. by Jeff Fort, New York: Zone Books, 2007.

⁶ Barthes, “The New Citroën,” p.90.

board, etc. – is only another outside of the same object, a folding of its exterior. Even here the relation to the car is dominated by the “entomological” smoothness of its body, the seamless transitions from metal to glass and vice versa. Hence the polished perfection of the dashboard: here too the rule applies that every gap, every fissure, and every hole that would remind the user of the object of the “technical and typically human operation of assembling” must be erased.⁷

It is this same smoothness and slickness that turns the *D.S.* into a medium. The new Citroën is not a thing filled with ‘metaphysical subtleties’ but a transmitter of news coming from a region beyond nature. As Barthes explains: “We must not forget that an object is the best messenger of a world above that of nature: one can easily see in an object at once a perfection and an absence of origin, a closure and a brilliance, a transformation of life into matter (matter is much more magical than life), and in a word a *silence* which belongs to the realm of fairy-tales.”⁸

Without any doubt we are confronted here with a leitmotiv in the writings of Barthes. The heavenly muteness of telling things fascinated him not only in *Mythologies*, but also in his discussion of Alain Robbe-Grillet’s novels (*Les gommes*) and Georges Bataille’s obscene narrations (*L’histoire de l’œil*), as well as in his study of Japanese culture.⁹ However, in 1950s France, Barthes was far from being the only author interested in the mythology of objects, and this collective interest paved the way for an ethnography of the present as it later was developed by Marc Augé, Bruno Latour and others.¹⁰

⁷ Ibid., p.88.

⁸ Ibid.

⁹ Roland Barthes, *Empire of Signs*, transl. by Richard Howard, New York: Hill and Wang, 1983.

¹⁰ Marc Augé, *Non-Places. Introduction to an Anthropology of Supermodernity*, transl. by John Howe, London etc.: Verso, 1995, id., *An Anthropology of Contemporaneous Worlds*, transl. by Amy Jacobs, Stanford, CA: Stanford University Press, 1999, and Bruno Latour, *We have never been modern*, transl. by Catherine Porter, Cambridge, MA: Harvard University Press, 1993.

In 1954, art historian Michel Carrouges published his famous investigation of “bachelor machines” in Marcel Duchamp and Franz Kafka, an explicit contribution to the study of the (machine-)mythologies created by modern societies. Two years later, another art historian, Pierre Francastel, finished a study of art and technology in the 19th and 20th centuries, largely inspired by Lewis Mumford and Siegfried Giedion, and which began with a chapter on the “myth of mechanization,” while already in 1951, economist Jean Fourastié had described in his *Machinisme et bien-être* the machine age as an authentic realization of the Faust myth, his main topic being the domination and transformation of nature by human technology.¹¹

Gilbert Simondon’s *Du mode d’existence des objets* also starts with a discussion of machine mythology.¹² Based on empirical studies of interior machine structures, Simondon criticizes the contemporary fascination with robots and other “mechanical slaves” as relying on an erroneous understanding of technological progress. According to Simondon, the decisive criterion for technological perfection is openness and flexibility rather than automation. The empirical character of this discussion is one of the reasons why, at least in the French context, Simondon’s book quickly became a “classical reference in the eyes of all of those who are studying our technological modernity.”¹³ In fact, authors such as diverse as

¹¹ Michel Carrouges, *Les machines célibataires*, Paris: Arcanes, 1954, Pierre Francastel, *Art et technique aux XIXe et XXe siècles*, Paris: Minuit, 1956, and Jean Fourastié, *Machinisme et bien-être*, Paris: Minuit, 1951.

¹² Gilbert Simondon, *Du mode d’existence des objets techniques*, Édition augmentée d’une préface de John Hart et d’une postface de Yves Deforge, 3rd ed., Paris, Aubier, 1989 (1st ed. 1958). In what follows I am quoting from the partial English translation by Ninian Mellamphy, published as typescript under the title *On the Mode of Existence of Technical Objects*, with a Preface by John Hart, University of Western Ontario, 1980. References are given in brackets with the short MEOT-E and the page number. In some cases, I have slightly modified this translation. Where further clarification was required, I co-quote the French original. Some quotations refer to parts of Simondon’s book that are not included in Mellamphy’s translation. In these cases, references are given in brackets with the short MEOT and the page number.

¹³ Gilbert Hottois, *Simondon et la philosophie de la „culture technique“*, Brussels: De Boeck, 1993, p.7. Up until today, the literature on Simondon has appeared primarily in the French-speaking countries. See, however, John

Herbert Marcuse, Jean Baudrillard, and Gilles Deleuze have made reference to Simondon's philosophy of technology and its terminological innovations, e.g. the process of "concretization" (i.e. the increase of functional over-determination inside the technical object), the issue of "hypertelia" (i.e. the phenomenon of overspecialization in technical objects that result in functional disadvantages) and the distinction between "major" and "minor uses" of technical objects (similar to Claude Lévi-Strauss's later distinction between engineer and *bricoleur*).¹⁴

Focusing on the introduction of *Du mode d'existence des objets techniques*, this essay explores Simondon's approach to technology by placing his seminal book in the discursive landscape of France in the late 1950s. It shows that Simondon's approach combines heterogeneous scientific and philosophical discourses on technology and the question of the object in a postwar consumer society into a unique mixture. It argues that Simondon, through his critical discussion of machine philosophy and cybernetics – from Pierre-Maxime Schuhl and Georges Canguilhem to Norbert Wiener – develops a highly original theory of technology *and* suggests an innovative form of politics aiming at better representing technical objects in contemporary society and culture.

The paper is divided into three sections. The first section deals with Simondon's critique of treatments of the machine as the "slave" of contemporary society. It shows that Simondon takes up a historical argument initially made by philosopher Pierre Maxime Schuhl in his *Machinisme et philosophie* (1938). Reedited in 1947 and reviewed by historian of science Alexandre Koyré, Schuhl's book highlights the problem of a social blockage of

Hart, „Preface“, in MEOT-E, pp.i-xxiii, and Paul Dumouchel, „Gilbert's Simondon's Plea for a Philosophy of Technology“, *Inquiry* 35, (3/4) (1992): 407-421.

¹⁴ Herbert Marcuse, *One-Dimensional Man. Studies in the Ideology of Advanced Industrial Society*, Boston: Beacon, 1964, Jean Baudrillard, *The System of Objects* [1968], transl. by James Benedict, London/New York: Verso, 1995, and Gilles Deleuze and Félix Guattari, *A Thousand Plateaus. Capitalism and Schizophrenia*, transl. by Brian Massumi, Minneapolis, MN: University of Minnesota Press, 1988. In addition, see Claude Lévi-Strauss, *The Savage Mind* [1962], transl. by George Weidenfield, Chicago: University of Chicago Press, 1968.

technological progress. Simondon argues that a similar blockage is about to occur in contemporary culture, especially if this culture continues to treat machines, in a rather conservative way, as a potentially threatening carrier of tools. Culture has to take into account the open and largely flexible nature of modern machines, and in particular the computer.

In the second section, the paper discusses Simondon's interpretation of machines as evolving entities characterized by internal conflicts in their arrangement of structures and functions. Studying empirical examples such as the combustion engine and the vacuum tube, Simondon literally opens the "black box" of technology, looking closely at the inside of concrete objects that he has disassembled mechanically and then reassembles by means of serial photographs. Although Simondon stresses the vitality of machines, he does not biologize them. In contrast to Ernst Kapp's theory of organ projection, for example, Simondon avoids directly referring technical objects to the human body, and while Georges Canguilhem highlights the importance of technique as a biological activity, Simondon sticks with the object as the decisive phenomenon of technology. According to him, technical objects are material entities that, with respect to the relation of human beings to nature, function as crucial mediators (*médiateurs*).

In the third section, the paper refers Simondon's views of our technological culture (or anti-culture) to Norbert Wiener's ideas about mass media and information technology. According to Wiener, information technology is a crucial "homeostatic factor" of all societies. In contrast, Simondon argues in favor of a small-scale approach to the problem of technology, in particular by focusing on human individuals and their local networks. According to Simondon, technical objects that have to be represented in our culture by human individuals are able to increase our technological knowledge and familiarity with machines.

1. Criticism

Not every discourse that concerns the myth of the machine leads to a discussion of technical objects. However, Simondon begins with this myth and, like Barthes, distinguishes two poles of cultural attitudes and behavior concerning technology. On one side he places a helpless admiration of technology as a “sacred object”, on the other, a non-reflective reduction of technology to mere everyday instrument. As a consequence, on the first pages of his book, Simondon presents the figure of the philosopher as a critic of “myths and stereotypes” (MEOT-E 7). This philosopher’s initial observation is that an imbalance or asymmetry exists in contemporary culture. Despite the fact that today’s society is thoroughly technological, it is characterized by the remarkable fact that it demarcates itself from technology. The common expression for this demarcation is the contrast or opposition between man and machine. This is a contrast that, according to Simondon, manifests itself in two forms: first, a reduction of machines to the status of simple devices or assemblages of matter that are constantly used but granted neither significance nor sense; second, and as a kind of response to the first attitude, there emerges an almost unlimited admiration for machines. Humans glorify technical objects as perfect automata or robots, and fascination for their features and performances tends to end in technology-centered utopias.

This double-sided demarcation of the technical from the human – technology seen as an entity or sphere that is remote from or above human beings – is in Simondon’s eyes reinforced by the fact that we only have limited cultural resources for consciously handling *things*. Only quite specific kinds of things are appreciated culturally as significant and precious, e.g. esthetic objects (paintings, novels, etc.) or religious objects (insignia, relics, etc.). This poverty of cultural competence with respect to things leads to confusion when we handle technical objects. Instead of recognizing them as such, i.e. as technical objects, one either takes them to be designer items or overestimates their worth, seeing them as authentic marvels and sacrosanct achievements.

What is lacking, then, is recognition and valuation of two seemingly opposed facts. On the one hand, we should take into account that tools and machines are made, driven and controlled by human beings. As Simondon puts it, the reality of technology is a genuinely human reality full of activity and productiveness (MEOT-E 1). On the other hand, we should be aware of the fact that technical objects have a ‘life of their own’, i.e. a specific mode of existence that can only be explored and defined in more detail if it is compared to the modes of existence of other objects (esthetic, religious, etc.) and other beings (human as well as non-human) (MEOT-E 12). Only by means of this double operation, which takes into account both continuities and discontinuities in the relation between human beings and technical beings, can we contribute to what Simondon sees as a completion and expansion of what is called, in contemporary societies, ‘culture’ and ‘humanism.’

Now the cultural asymmetry that the philosopher diagnoses at the beginning of his study is also an a-synchronicity (*Ungleichzeitigkeit*). Both ignoring and admiring technology demonstrates, in Simondon’s view, that in the very heart of modernity an archaic pattern perpetuates itself. According to him, the society of the 20th century continues to comport itself toward technical objects as if it were a society of the 19th century. Its schemes of thinking and acting remain tied to craft and agriculture and have never really adapted to cities and industries. This is for this reason that a narrow-minded antipathy toward technology exists alongside a futuristic fascination for technology. From the perspective of this antipathy, technical objects manifest themselves as barbarian invaders: “Culture behaves toward the technical object much in the same way as a man caught up in primitive xenophobia behaves toward a stranger” (MEOT-E 1). Machines have status of foreign workers. We rely on them and their service, we know their names and types, but do not want to learn anything more specific about them. We do not wish to have closer relations to those beings, let alone enter into “dialogues” with them. We’re not sure: is there any common language available for such a dialogue? We feel comfortable stressing the need for closer integration of machines into our

human patterns of living. But we have a hard time facilitating such integration in concrete and practical ways.

Simondon goes even further. In the very first paragraph of his study he portrays machines as the slaves of modern society:

The purpose of this study is to attempt to stimulate awareness of the significance [*sens*] of technical objects. Culture has become a system of defense designed to safeguard man from technics. This is the result of the assumption that technical objects contain no human reality. We should like to show that culture fails to take into account that in technical reality there is a human reality, and that, if it is fully to play its role, culture must come to terms with technical entities [*les êtres techniques*] as part of its body of knowledge and values. Recognition of the modes of existence of technical objects must be the result of philosophic consideration; what philosophy has to achieve in this respect is analogous to what the abolition of slavery achieved in affirming the worth of the individual human being (MEOT-E 1).

The provocative nature of this passage is intentional. As in sci-fi novels, traditional culture criticism has often described man as ‘slave to the machine.’ Simondon inverts this figure. It is not man who is slave to the machine; rather, technical beings have become subject, as slaves, to human beings.

The comparison of humans and technical beings which serves as the basis of this argument may seem exaggerated. In place of humanism, Simondon substitutes only an expansion of anthropomorphism. However, he could rely on and refer to contemporary discourses that actually described automata and robots as “mechanical slaves.” According to the cybernetically inspired techno-enthusiast Albert Ducrocq, for example, increasing

exploitation of these slaves will contribute to solving, finally, basic human problems such as work, health, security, etc.¹⁵

The parallel that Simondon draws is more convincing, moreover, if we recall that slavery in antiquity went hand in hand with the loss of all rights that were otherwise granted by birth and/or origin. Precisely because this is also the case for machines, Simondon seeks to reconstruct the emergence and evolution of technical objects. At this point in his text, then, his interest in the emergence and evolution of machines appears as a first step toward integrating technical slaves into our culture.

More concretely, Simondon's claim that machines are slaves resonated with contemporary discussions in French philosophy and history of science. Since the late 1930s, philosopher Pierre-Maxime Schuhl had argued that slavery in antiquity had been a decisive obstacle to a more focused development and use of technology in that period. In *Machinisme et philosophie*, slavery functioned as a synonym for a "mental blockage" with respect to technology, an idea that Schuhl reiterated in 1947, when publishing the second edition of his study. He explained in this latter edition that, in classic Greece: "If one did not make recourse to machines on a large scale, it was because there was no need to reduce manual labor, given the fact that one had (inexpensive and numerous) living machines ready at hand who were as distanced from the free man as animals, i.e., slaves."¹⁶

Shortly afterward, historian of science Alexandre Koyré discussed and modified this "socio-psychological argument" and expressed his skepticism with respect to any attempt at "explaining" historically the specificities of the culture of technology in ancient Greece.¹⁷

However, in his extended review of *Machinisme et philosophie*, Koyré reminded his readers

¹⁵ Albert Ducrocq, *L'ère des robots*, Paris: Julliard, 1953, p.10.

¹⁶ Pierre-Maxime Schuhl, *Machinisme et philosophie*, 2nd rev. ed., Paris: PUF, 1947, p.10.

¹⁷ Alexandre Koyré, "Les philosophes et la machine" [1948], in: id., *Etudes d'histoire de la pensée philosophique* Paris: Gallimard, 1971, p.305-339, on p.336. On philosophy of technology in France, see Daniel Cérézuelle, „Fear and Insight in French Philosophy of Technology,“ *Research in Philosophy and Technology* 2 (1979): 53-75, and Jean-Yves Goffi, *La philosophie de la technique*, Paris: PUF, 1988.

of the fact that one of the characteristic features of Aristotle's philosophy lies in identifying "animate" and "inanimate tools." A passage in his *Politics*, for example, can be literally read as cybernetic: "For a helmsman the rudder is a lifeless tool and the look-out man a live tool—for an assistant in the arts belongs to the class of tools."¹⁸

At the beginning of his study, Simondon returns to this relation of replacement between tool and assistant. His discussion does not evoke the question of cybernetics, however. Rather, it addresses the political theme of culturally rehabilitating machines and stimulating awareness of their significance and value. As a result, when Simondon portrays the machines of our times as "slaves", we can also read this as referring to the danger of technical and social stagnation in the near future, a danger that expresses itself in contemporary societies as a cultural blockage with respect to technology. This blockage translates itself into that non-reflective use of technical objects that dominates our everyday experience. At the same time, this blockage leads to a misconception of what technical progress means, a misconception that is transmitted by the mythological image of the robot.

According to Simondon, it is precisely *not* an increasing automation (=enslavement) of technical objects that is characteristic of technological improvement. Rather, it is in the emergence of machines that can be used in multiple ways – machines with "margins of indetermination" that allow for flexible couplings with other machines and humans – that we can observe the highest degree of technicity. As we will see in the third section of this paper, it is only against the background of this insight into the openness and controllability of technology that Simondon's project of machine-slave-liberation as a means for integrating technical objects into culture becomes plausible.

2. Interpretation

¹⁸ Aristotle, *Politics*, with an English translation by H. Rackham, Cambridge, MA/London: Harvard University Press, 1990 (=Loeb Classical Library), p.17 (1253b 28-31).

After playing the part of the critic, Simondon's philosopher takes up a second role. This is the role of the interpreter, i.e. the analyst of forces and their relation. The 'significance' or 'sense' (*sens*) of technology of which he tries to make us aware is not the 'meaning' of a linguistic entity that could be determined by means of semantics or hermeneutics. Rather, sense is to be understood as a dynamic relation between potentials that are literally realized in the technical object and develop there by means of inner resonances. Here, the originality of Simondon's project becomes perhaps especially apparent. The sense of technology is not derived from any kind of exterior reference to it, whether mediated by word, image or hand. Rather, it can be explored and inferred by reconstructing the history of force relations *inside the technical object*.

For Heidegger, the 'thingyness' of things is described in terms of a cavity, an empty space, in which heaven and earth, immortals and mortals gather. In cybernetics, the interior of the object appears as similarly dark, as the unobservable content of a black box.¹⁹ Simondon, by contrast, assumes a kind of transparent fullness. He quite concretely looks at the potentials and relations that are condensed in the technical object, and which concretize themselves by means of "functional over-determination"—and, by the same token, confer "consistence" on this object so that, as a result, the object can be seen as a specific "mode of existence" (MEOT-E 8). From this perspective, the technical object presents itself as a "theater of a number of relationships of reciprocal causality" (MEOT-E 22). It is a scene of structural and functional conflicts that are solved gradually or suddenly. The overarching goal of the concretization process is a state in which the object is "no longer divided against itself"

¹⁹ See Martin Heidegger, "The Thing," in: id. *Poetry, Language, Thought*, transl. by Albert Hofstadter, New York: Harper & Row, 1971, p.165-186, and W. Ross Ahsby, "The Black Box," in: id., *An Introduction to Cybernetics*, London: Chapman & Hall, 1956, p.86-117.

(MEOT-E 30), or, to be more precise, no longer fights with itself (*n'est plus en lutte avec lui-même*) (MEOT 34).²⁰

This means, conversely, that 'sense' is not to be understood as a simple biological phenomenon that could be deduced from the position of tools and machines in the physiological economy of the human organism. In contrast to Georges Canguilhem, technology for Simondon is not a manifestation of an activity that is "rooted in the living being's spontaneous effort to dominate the environment and organize it according to his values as a living being."²¹ Simondon more cautiously only speaks of technical objects as "mediators" (*médiateurs*) between nature and humans. As in the case of Barthes, Simondon's study of the technical object prolongs itself into a discourse on media. However, where Barthes was, in this context, interested in the phenomenon of communication, Simondon orients his media theory of technology towards the problem of labor.

This becomes evident in his discussion of the "hyle-morphic model" of substance, i.e. the ancient idea that bodies in the widest sense result from imprinting a form (*morphè*) to matter (*hylè*). With respect to those activities that result in the emergence of technical objects, Simondon contests the validity of the hyle-morphic model, arguing that this model itself relies on a technical analogy, namely the fabrication of bricks. He opposes the image of the sculptor who, according to the hyle-morphic model, simply imposes his vision onto the marble block upon which he is working. Implicitly, Simondon then argues against the idea of the worker who has his product "in mind" before he produces it in reality and the scientist who understands technology as the mere result of applying well-established laws to given problems. Instead, he presents the creator of technical objects as acting in a region beneath

²⁰ More generally, see Bernard Stiegler, "The Theater of Individuation. Phase-Shift and Resolution in Simondon and Heidegger", transl. by Kristina Lebedeva, *Parrhesia* 7 (2009): 46-57.

²¹ Georges Canguilhem, *The Normal and the Pathological*, transl. by Carolyn R. Fawcett, New York: Zone, 1989, p.228-229. See also his seminal article "Machine and Organism" [1952], in Georges Canguilhem, *Knowledge of Life*, transl. by Stefanos Geroulanos and Daniela Ginsburg, New York: Fordham University Press, 2008, p.75-97.

morphè and *hylé*, a region in which forms are materialized and, simultaneously, matter is rendered formable. For this inter-actor, nature is not a homogeneous substance that can be fully controlled by artificial means, but is rather a mixed and multiple materiality, often filled with resistances, that transports a multiplicity of “implicit forms.”²² The machine maker has to cooperate with these matter-forms and form-matters if he would like to arrive at a new technical object. He places himself in this pre-figured and pre-fabricated realm, and moves within its tracks so that from them he can create an object.

This conception differs clearly from the theory of organ projection that Ernst Kapp had introduced into the philosophy of technology in the late 19th century. Kapp actually coined the phrase “philosophy of technology,” and his argument concerning the creation of tools by exteriorizing organic forms, and in particular forms derived from the hand, remained highly attractive until at least the 1950s.²³ With respect to tools such as the pincer and the hammer, Simondon similarly claims that they “extend as well as amplify and protect” isolated parts of the human organism.²⁴ However, his argument is not as causally-oriented as Kapp’s. Rather, Simondon prefers a description of functional connections, and in particular “dynamic schemes” (MEOT-E 7) of *gestures*. Yet even when he does argue for a slightly more causal connection, his image is different from that of Kapp; for Simondon, “Human reality resides in machines as human actions fixed and *crystallized* in functioning structures” (MEOT-E 4, my emphasis).

²² Gilbert Simondon, *L’individuation à la lumière des notions de forme et d’information*, Grenoble: Millon, 2005, p.55.

²³ Ernst Kapp, *Grundlinien einer Philosophie der Technik. Zur Entstehungsgeschichte der Kultur aus neuen Gesichtspunkten*, Braunschweig: Westermann, 1877. On Kapp, see Carl Mitcham, *Thinking through Technology. The Path between Engineering and Philosophy*, Chicago: University of Chicago Press, 1994, p.20-24, and Grégoire Chamayou, „Présentation“, in: Ernst Kapp, *Principes d’une philosophie de la technique*, Paris: Vrin, 2007, p.7-44.

²⁴ Simondon, *L’individuation*, p.51.

This image of crystallization refers back to Simondon's important study of the individual and its physical-biological genesis. In that study, Simondon's principal doctoral thesis, crystallization is the base analogy for the process of individuation of matter. Thus, Simondon's conception of the emergence of tools refers not to the lived body (as in Kapp), but to matter itself and to its crystallization in human as well as non-human forms.

There is another twist by means of which Simondon distances himself from a projection theory of technology. With respect to the body-tool relation that he highlights, the effect of prolongation is not one-dimensional, and this is true even and especially in tools. As Simondon points out, and in keeping with theme of mediation noted (and quoted) above, the effect of prolongation leads back as well from the tool to the body: "Knowing how to use a tool does not just mean having acquired the necessary gestures; it also means that one is able, by means of the signals that come through the tool to the human being, to recognize the implicit form of matter, exactly at the point where the tool is applied."²⁵ In this sense, a planer, for example, is not simply a device that lifts off a more or less substantial strip of wood. It is, at the same time, a means that makes it possible to sense whether or not the strip can be removed smoothly, without producing splinters, or, if the movement suddenly feels harsh, is a movement in which one no longer is moving along the grain.²⁶ To a certain degree, then, what holds true for the creator of technical objects also applies to the user of such objects: The carpenter does not deal with a raw and abstract stuff onto which he can impose his scheme by means of any given organ projection, but he is rather handling a 'matter of signs' that he has to follow – not once and for all, but again and again – during the realization of his plan by means of the technical object that he employs.²⁷

²⁵ Ibid., p. 53.

²⁶ Ibid.

²⁷ On this point, see also Brian Massumi, *A User's Guide to Capitalism and Schizophrenia. Derivations from Deleuze and Guattari*, Cambridge, MA etc.: MIT Press, 1992, p.10.

3. Politics

After playing the critic and the interpreter, Simondon's philosopher again enters the scene, this time in a third role. One could speak of this role as that of future politician garbed in an ancient robe. Despite his refusal of the hyle-morphic model, Simondon makes clear in his study of technology as a whole that he reconnects his own reflection to an Aristotelian ontology that, at least initially, does not postulate any radical difference between natural and artificial products and/or processes. As we have already seen, this reconnection with Aristotelian thought is one of the preconditions for the comparison that Simondon draws between machine and slave, a comparison which, in turn, represents a rather striking starting point for elaborating the specific mode of existence of technical objects. At the same time, however, Simondon argues that it is not Aristotelian philosophy that opens up a concrete perspective for extending and completing culture, but rather the latest state of technology. Philosophy is able to criticize myths and stereotypes, and it can contribute, by means of retracing the interior force relations of technical objects, to an understanding of the sense of these mediating objects. However, these tasks are only of value, "both politically and socially" (MEOT-E 7), when philosophy actively enters the transitional zone between culture and technology. It is the notion of regulation that mediates this move.

For Simondon, culture is the "basis of meanings, modes of expression, proofs and forms." It is a system of "schemes, symbols, qualities, analogies" that has regulatory effects in the broadest sense. Culture emerges from the "life of the group" and from there it is transferred to the "gestures" of those who carry out group-leading functions, while the group defines the "schemes" of these gestures and attributes "norms" to them (MEOT-E 6-7). On the one hand, culture appears as that which enables any governance at all; on the other hand, culture relies on a delegation of this necessary power from those who govern to those who are governed and vice versa (MEOT 150). As a consequence, the functioning of culture can be

described in its basic outline in the following way: “a culture establishes regulatory communication among those who share that culture” (MEOT-E 6).

Using the terminology of cybernetics, one might say then that culture is a “homeostatic factor” in the life of societies. Insofar as culture is understood as a homeostatic factor, it occupies the precise position that Norbert Wiener assigned to news services and, more generally, all means of communication. However, Wiener was skeptical as to whether or not the use of communication means, even if it took place with “intelligence, skill, and honesty of purpose,” would be able to oppose the forces of “buying and selling” that, according to him, were basically “anti-homeostatic” and which still dominated modern societies.²⁸ Moreover, Wiener was quite pessimistic that individuals or small groups would be able to enhance the homeostatic functions of modern societies. According to him, this was a reasoning relying on “the mode of thought of the mice when faced with the problem of belling the cat. Undoubtedly it would be very pleasant for us mice if the predatory cats of this world were to be belled, but – who is going to do it?”²⁹ Simondon thinks differently and refers to the “bitterness” of Wiener’s view (MEOT 150). Unlike the author of *Cybernetics*, the French philosopher puts trust in the balancing powers of the community, i.e. culture in the sense of regulative connections (*Bindung*) and knowledge (*Bildung*).

According to Simondon, the present development of technology is characterized by assembling the machines in totalities (*ensembles*) that permit and require regulations. *De facto*, such ensembles are already part of culture, and it seems all the more bizarre that, *de jure*, they are still excluded from it. It is precisely this kind of a transitional state of affairs that prepares, in Simondon’s eyes, the transition to a culture of technology: “Since technical reality has become regulatory, it can be integrated into culture, which is itself essentially regulatory” (MEOT-E 9).

²⁸ Norbert Wiener, *Cybernetics. Or Control and Communication in the Animal and the Machine*, New York/Paris: John Wiley/Hermann, 1948, p.190 and p.188.

²⁹ *Ibid.*, p.189.

He then quotes two examples in order to illustrate this ‘becoming regulatory’ of technology. The first example is provided by “modern calculating machines.” These machines contain “a very great range of circuit-commutations which make it possible to program the working of the machine, e.g. to work out cubic roots or to translate from one language into another a simple text” (MEOT-E 5). This “openness” or “freedom in functioning” shows that calculating machines are not “pure automata” but technical beings with which human beings can cooperate in authentic ways.

It is a similar “margin of indetermination” that allows for assembling machines to become part of “coherent ensembles” within which they can mutually exchange information. At this point, Simondon introduces his second example, i.e. the radio. In this case, too, he insists on the role of interaction and translation. Even if a direct exchange of information takes place between two machines – “such as between a pilot oscillator and another oscillator synchronized by impulses” – the human individual also intervenes “as a being who regulates the margin of indetermination so as to make it adaptable to the greatest possible exchange of information” (MEOT-E 5). Summarizing both examples, Simondon then adds that “the ensemble of open machines assumes man as permanent organizer and as a living interpreter of the interrelationships of machines” (MEOT-E 5).

Simondon suggests that the consequences of accepting these claims are far reaching. Instead of continuing to conceive of himself as an individual who is confronted with isolated technical objects and who feels threatened or redeemed by them, man can position himself in a completely differently way. He becomes a witness and interpreter of the difficulties of technical objects and a mediator of their mutual relations. He starts to act as “a sociologist or psychologist of machines” (MEOT-E 6) who lives right among them:

Far from being the supervisor of a squad of slaves, man is the permanent organizer of a society of technical objects which need him as much as

musicians in an orchestra need a conductor. The conductor can direct his musicians only because, like them, and with a similar intensity, he can interpret the piece of the music performed; he determines the tempo of their performance, but as he does so his interpretative decisions are affected by the actual performance of the musicians; in fact, it is through him that the members of the orchestra affect each other's interpretation; for each of them he is the real, inspiring form of the group's existence as group; he is the central focus of interpretation of all of them in relation to each other. This is how man functions as permanent inventor and coordinator of the machines around him. He is *among* the machines that work with him (MEOT-E 4).

The image of the democratic orchestra is highly evocative. In such an interaction with machines, based on margins of indeterminacy and sophisticated tuning, Simondon recognizes a genuinely cultural task, not just a matter for economists and/or engineers. As a result, it is not the worker in the factory or someone in an office working with a *single* machine who is prepared and able to understand this challenge in its cultural dimensions, nor is it the owner of a factory or its manager who submits the functioning of a technical ensemble to *one* specific purpose, i.e. capitalist profit. At the same time, however, the man-machine-concert should not be understood in the sense of a cultural utopia that would prolong the use of calculating machines and radio into the future.

If one was to be affected by what Canguilhem calls the "virus of the precursor," one might be tempted to see in Simondon's bringing together of these two technologies an "anticipation" of the internet. However, this is precisely what it is not. Rather, Simondon's discussion of culture and technology, circa 1951, is meant to improve our abilities to perceive, understand, and interact with contemporary technical ensembles such as laboratories, factories, or centers of calculation. However, what these ensembles represent is not just a

technical but also a social potential. It is question here of “societies” of technical objects that overlap and interact in multiple ways with human societies. This idea is highlighted in the image of the orchestra because the musicians who stand in for the machines are themselves complex assemblages of body and instrument.

Conclusion

The difference between Simondon and Wiener is obvious. It is true that Simondon acknowledged cybernetics as a contribution to the general phenomenology of machines and recognized in it a new approach to the inductive study of technical objects.³⁰ However, he is quite critical of Wiener’s fascination with a specific type of machine, i.e. feedback mechanisms, and argues against what he sees as Wiener’s over-arching tendency to compare technical with natural objects without also seeking to highlight specific differences between the two (MEOT-E 49-50). Simondon raises similar objections to Wiener’s claims about the way in which social homeostasis functions in the case of communication means and news services. In his eyes, Wiener’s approach runs the danger of reproducing a separation between culture and technology. In a sense, Wiener attempts to solve the regulation problem technologically, instead of embedding tools and techniques for regulation in cultural processes.

By contrast, what Simondon seeks are not new media and communication technologies, but individuals who would be able to describe and explain the orchestra-like interaction between human beings and technical beings. More concretely, he is in search of persons who, on the one hand, can position themselves at some remove from the individualization of technology yet who also would have, on the other hand, enough “technical wisdom” (MEOT 148) that they would be able to function as authentic spokesmen

³⁰ On cybernetics and phenomenology see also François Russo, „La cybernetique située dans une phénoménologie générale des machines,“ *Thales* 7 (1953): 69-75.

in favor of machines. The principal task of these experts would consist in operating as “representatives of technical beings” (MEOT 151), in particular vis-à-vis those humans whom we believe contribute essentially to our cultural life: writers, artists, and other especially visible personalities.

If one takes Simondon’s allusions to cultural democracy seriously, then a large number of these “representatives of technical beings” would be required. As advocates of culture *and* technology, they would by no means be lobbyists acting in the name of some machine industry, but rather ambassadors of individual technical objects who, like political representatives, would facilitate acceptance and respect for these objects as significant components of cultural life.

Simondon leaves hardly any doubt that this kind of political activity would not lead to great popularity. This is, at least, how one can read an interview he gave in the early 1980s. By this point, the automobile had become his prime example for the cultural necessity of liberating and “saving” the technical object, both from an increasingly careless means of production as well as from a degrading kind of use.³¹ Even in his 1958 book, Simondon appeared to be rather skeptical about the technological significance of cars. According to him, the automation of the window crank and the introduction of servo steering in fact represented technological complications, though car manufacturers and salesmen praise them as simplifications. More generally, he states: “The automobile, this technical object that is so charged with psychic and social implications, is not suitable for technical progress.” (MEOT-E 21).

In 1983, Simondon came back to this normative criticism of the technical object, underscoring that the use of cars continues to be focused too much on its non-technical aspects: “I have seen a commercial that praised the footboards of a particular car. Such a

³¹ Gilbert Simondon, “Sauver l’objet technique (Entretien avec Anita Kechikian),” *Esprit* 7/4 (1983): 147-152, on p.148.

beautification of the technical object by means of things other than its technicity has to be refused.”³² In principle, he continues, there is nothing about the aesthetization and/or erotization of technical objects to which we should object. However, such tendencies should refer to “margins of indetermination,” i.e. the technical openness of the object. The result is a kind of Bauhaus vision of the car that is in striking contrast to Barthes’s elegant appraisal of the new Citroën: “From the aesthetical and technical point of view, a truck seems to me purer than a simple car.”³³

Along with cars, today’s culture has become saturated with digital media technology. However, even in the age of ubiquitous computing, Simondon’s theory and politics of technology remains extremely relevant and instructive. It continues to confront us with a double task. First, there is a persisting need of stimulating and developing awareness of the cultural significance and value of technological objects. While today we are experiencing a growing interest in Google, Facebook, and Twitter and their impact on societal phenomena – from democracy to issues of authorship –, only little interest is directed towards the technical objects and ensembles that makes possible their use. In this situation, Simondon’s theory of technology asks us to draw our attention to the materiality of technological objects and ensembles that enable us to use these digital platforms: servers, cable networks, radio masts etc. It suggests developing an informed awareness of these largely invisible infrastructures and urges to accept and respect them as crucial elements of our culture.

The second task consists in the normative criticism of technological objects. The main issue in this regard is the problem of regulation. Simondon insists on this point: The principal feature of technological improvement is not an increase in automation but the emergence and evolution of machines that are open to regulation. In this perspective, digital technologies with visually attractive yet ready made and black-boxed interfaces appear as highly

³² Ibid.

³³ Ibid., p.149.

problematic. These technologies derive their popularity from a specific kind of aesthetics, not because they provide us with enough “margins of indetermination”. Similar problems are embodied in the vast array of technological objects that populate our everyday life: from refrigerators and washing machines to coffee makers. Simondon urges us to look inside these black boxes. This is why his philosophy of technology is an important contribution to understanding and shaping the material culture of contemporary societies.